Abstract

Association between ambient temperature exposure and birth outcomes: The role of season of birth in Taiwan

Zi-Yan Wang[1], Wan-Chen Lin[1], Ming-Lun Zou[1], Shih-Chun Candice Lung[2], Hsing Jasmine Chao[1] and **Yi-Hua Chen**[1]
(1)Taipei Medical University, New Taipei, Taiwan, (2)Academia Sinica, Taipei, Taiwan

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BACKGROUND: Temperature exposure during pregnancy was found to affect birth outcomes. However, in addition to inconsistent results, geographical differences in the context of global climate change should be of particular concern. Therefore, our study aims to examine the association between ambient temperature exposure during each trimester and birth weight, with an emphasis on the effects of the season of birth in subtropical Taiwan.

METHODS: Participants were selected from an ongoing cohort study: the Longitudinal Examination across Prenatal and Postpartum Health in Taiwan (LEAPP-HIT). We analyzed the data from December 2011 to November 2022, with 1,245 participants included for examination. Average (AT), maximum (MaxT), minimum (MinT), and range (MaxT minus MinT, RT) of ambient temperature were assessed using data from the Central Weather Administration. Both linear regression models and logistic regression models were employed, with subgroup analysis conducted by season of childbirth, including cold months (November-March) and hot months (June-September).

RESULTS: We found that higher MinT during the third trimester was independently associated with higher risks of low birthweight (LBW) and lower birth weight, while a wider RT during the third trimester was associated with lower risks of LBW and higher birth weight. Seasonal disparities were particularly observed. Specifically, for children born in the cold season, higher AT and MaxT during the third trimester were associated with lower risks of LBW ($OR=0.73$, $p=0.04$ and $OR=0.70$, $p=0.001$, respectively) and higher birth weights ($\beta=23.78$, $p=0.03$ and $\beta=21.73$, $p=0.02$, respectively). Nevertheless, for children born in the hot season, we found that higher AT ($\beta=-25.61$, $p=0.004$) and MinT ($\beta=-19.25$, $p<0.001$) during the third trimester were associated with lower birth weights.

CONCLUSIONS: Our findings highlight the importance of prenatal ambient temperature exposure on birth outcomes, particularly concerning seasonal effects, and stress the need for pregnant women to receive adequate monitoring in light of climate change.

Environmental health sciences Epidemiology Other professions or practice related to public health Public health or related public policy Public health or related research Social and behavioral sciences